WHEN THE IDEAL IS THE WHOLE RING THE GLUED CATEGORY IS EQUIVALENT TO THE CATEGORY OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 24767-24774 (canonical tag 05EP) and the counted proof is the proof environment at source lines 24776-24789, which immediately follows it and is its own proof.

Editorial note. The statement says that the ideal generated by the elements equals the whole ring, which is the degenerate case of the equivalence proved later in the section; the two lemmas cited are an algebra result on glueing modules and the earlier statement that one composite is the identity, and both kinds recorded are the type words appearing in their labels.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note, and the preamble span runs to line 90, the current end of the retained preamble.

Editorial note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- Glueing data consisting of modules over the localizations with compatible transition isomorphisms come from a unique module whose localizations realize the given modules. invoked at proof lines 24776-24780. Produces the module whose image under the functor can be compared with the given glued object.
- Two modules over the flat cover that become isomorphic after localizing at each of the elements are canonically isomorphic. used at proof lines 24780-24784. Identifies the first component of the glued object with the base change of the module produced by glueing.
- The composite of the functor with the functor back is isomorphic to the identity on modules. invoked at proof lines 24786-24788. Together with the essential image statement, gives the equivalence of categories.

Source reference. Original lemma 00EQ: sources/stacks-chow/tags.txt, source lines 544–544. The reference supplies the glueing of modules over a flat cover used at proof lines 24776-24780; the comparison of the components that follows is inside the counted span.

Source reference. Original lemma 05EM: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24729–24729. The reference is cited at proof lines 24786-24788 to combine with the essential image statement and obtain the equivalence; it is the last step and the rest of the proof is inside the counted span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $\varphi : R \to S$ be a flat ring map and $(f_1, \ldots, f_t) = R$. Then Can and H^0 are quasi-inverse equivalences of categories*

$$Mod_R = Glue(R \to S, f_1, \ldots, f_t)$$

Proof. Consider an object $\mathbf{M} = (M', M_i, \alpha_i, \alpha_{ij})$ of Glue$(R \to S, f_1, \ldots, f_t)$. By Algebra, Lemma 00EQ there exists a unique module M and isomorphisms $M_{f_i} \to M_i$ which recover the glueing data α_{ij}. Then both M' and $M \otimes_R S$ are S-modules which recover the modules $M_i \otimes_R S$ upon localizing at f_i. Whence there is a canonical isomorphism $M \otimes_R S \to M'$. This shows that $\mathbf{M}$ is in the essential image of Can. Combined with Lemma 05EM the lemma follows. □